Ordinary Differential Equations

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Contents

1 Introduction to Ordinary Differential Equations **2**

2 Falling Object Model **2**

2.1 Physical Modeling and Forces 2

2.2 Numerical Example & Direction Field Analysis 3

2.3 Qualitative Behavior Conclusions 4

2.4 General Algebraic Conclusions 4

3 Population Model **4**

3.1 Scenario A: Exponential Growth (No Predation) 5

3.2 Scenario B: Introducing Predation (Owls) 5

3.3 Quantitative Analysis and Slopes 5

1 Introduction to Ordinary Differential Equations

Differential equations are mathematical equations that relate a function with its derivatives. In physical applications, the independent variable often represents time (t) or space (x), and the dependent variable represents a changing quantity (y or v).

Here are a few introductory examples of explicit functions and the differential equations they satisfy:

$$y = x^2 \implies \frac{dy}{dx} = 2x = f(x)$$

$$y = e^x \implies \frac{dy}{dx} = y = f(y)$$

$$x^2 + y^2 = R^2 \implies \frac{dy}{dx} = -\frac{x}{y} = f(x, y)$$

In general, a first-order ordinary differential equation (ODE) can be written in the form:

$$\frac{dy}{dx} = h(x, y) \quad \text{or} \quad \frac{dy}{dt} = h(t, y)$$

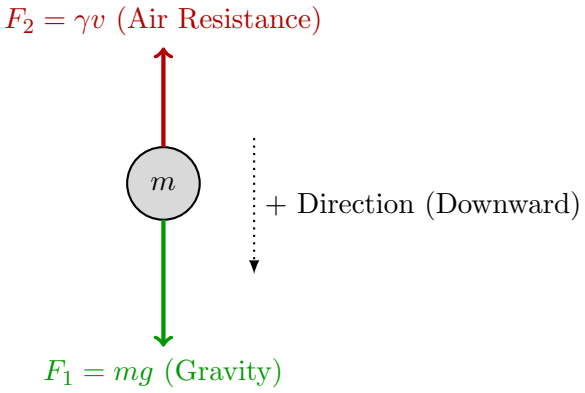
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2 Falling Object Model

Consider an object of mass m falling through the atmosphere near the Earth's surface. We model the forces acting on the object to understand its velocity $v(t)$ over time.

2.1 Physical Modeling and Forces

There are two primary forces acting on the object: gravity pulling it downward and air resistance (drag) opposing the motion upward.



Where:

- m, g, and γ are positive constant parameters.
- $g = 9.81$ m/sec^2 represents acceleration due to gravity.
- γ represents the drag coefficient (resistance parameter).

Applying Newton's Second Law ($F = ma$):

$$ma = F_1 - F_2$$

Since velocity is $v = v(t)$ and acceleration is $a = a(t) = \frac{dv}{dt}$, we can substitute these into the force balance:

$$m\frac{dv}{dt} = mg - \gamma v$$

Dividing through by the mass m yields our explicit first-order linear ODE for velocity:

$$\frac{dv}{dt} = g - \frac{\gamma}{m}v$$

2.2 Numerical Example & Direction Field Analysis

Let us assign specific numbers to the parameters:

- Mass $m = 10$ kg
- Drag coefficient $\gamma = 2$ kg/sec
- Acceleration due to gravity $g = 9.8$ m/sec^2

The equation becomes:

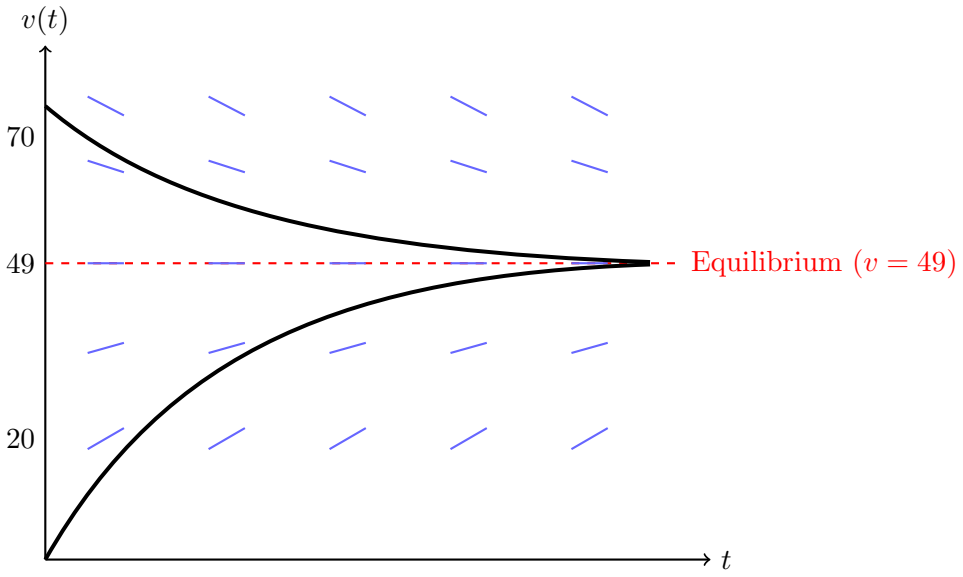
$$\frac{dv}{dt} = 9.8 - \frac{v}{5}$$

Assuming the object is dropped from rest, we have the initial condition: $v(0) = 0$.

By evaluating the derivative $\frac{dv}{dt}$ at various values of v, we can track how the slope of our solution changes:

- If $v = 0 \implies \frac{dv}{dt} = 9.8$
- If $v = 10 \implies \frac{dv}{dt} = 7.8$
- If $v = 40 \implies \frac{dv}{dt} = 1.8$
- If $v = 49 \implies \frac{dv}{dt} = 0$
- If $v = 60 \implies \frac{dv}{dt} = -2.2$
- If $v = 80 \implies \frac{dv}{dt} = -6.2$

Visualizing these slopes creates a **direction field** (or slope field). Below is a sketch of the direction field along with the equilibrium solutions:



2.3 Qualitative Behavior Conclusions

From this directional analysis, we observe that as $t \to \infty$, the velocity $v(t) \to 49$ m/sec.

1. If $v(0) = 49$, then $v(t) = 49$ for all t (constant solution).
2. If $v(0) < 49$, then $v(t) \to 49$ as $t \to \infty$ in a strictly increasing fashion.
3. If $v(0) > 49$, then $v(t) \to 49$ as $t \to \infty$ in a strictly decreasing fashion.

2.4 General Algebraic Conclusions

Returning to the algebraic equation $\frac{dv}{dt} = g - \frac{\gamma}{m}v$, setting $\frac{dv}{dt} = 0$ yields the **equilibrium velocity** (also known as the limiting or terminal velocity):

$$v_e = \frac{mg}{\gamma}$$

1. If $v(0) < \frac{mg}{\gamma}$, then $v(t) \to \frac{mg}{\gamma}$ as $t \to \infty$ in an increasing way.
2. If $v(0) > \frac{mg}{\gamma}$, then $v(t) \to \frac{mg}{\gamma}$ as $t \to \infty$ in a decreasing way.
3. If $v(0) = \frac{mg}{\gamma}$, then $v(t) = \frac{mg}{\gamma}$ is a constant equilibrium solution.

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3 Population Model

Let $p(t)$ represent the population size of field mice at time t.

3.1 Scenario A: Exponential Growth (No Predation)

Assuming ideal environmental conditions with no predators, the population grows at a rate directly proportional to its current size:

$$\frac{dp}{dt} = rp$$

Where $r > 0$ represents the constant intrinsic growth rate parameter.

3.2 Scenario B: Introducing Predation (Owls)

Now assume a population of owls relocates to the area and kills/eats a steady number of mice, k, per unit of time. The rate equation shifts to account for this constant loss:

$$\frac{dp}{dt} = rp - k$$

3.3 Quantitative Analysis and Slopes

Let's analyze a specific case where the intrinsic growth rate $r = \frac{1}{2}$ and the predation rate $k = 450$ mice/month. Our differential equation becomes:

$$\frac{dp}{dt} = \frac{1}{2}p - 450$$

To evaluate the long-term behavior of the mouse population, we find the equilibrium point by setting $\frac{dp}{dt} = 0$:

$$\frac{1}{2}p - 450 = 0 \implies p_e = \frac{k}{r} = \frac{450}{1/2} = 900 \text{ mice}$$

Long-Term Dynamic Behaviors:

- **Thriving Population ($p(0) > \frac{k}{r}$):** If the starting population is greater than 900, the reproductive rate outpaces predation. The growth rate $\frac{dp}{dt}$ stays positive and expands, meaning $p(t) \to \infty$ as $t \to \infty$.
- **Extinction Event ($p(0) < \frac{k}{r}$):** If the starting population is fewer than 900, the owls consume mice faster than they can reproduce. The growth rate $\frac{dp}{dt}$ is negative, meaning $p(t) \to 0$ in finite time (mathematically heading towards $-\infty$).
- **Perfect Balance ($p(0) = \frac{k}{r}$):** If the starting population is exactly 900, the birth rate perfectly counters the predation rate. The population remains static at $p(t) = 900$.